

MIAMA active travel and health impact methodology

Draft methods description for methodological review and early testing

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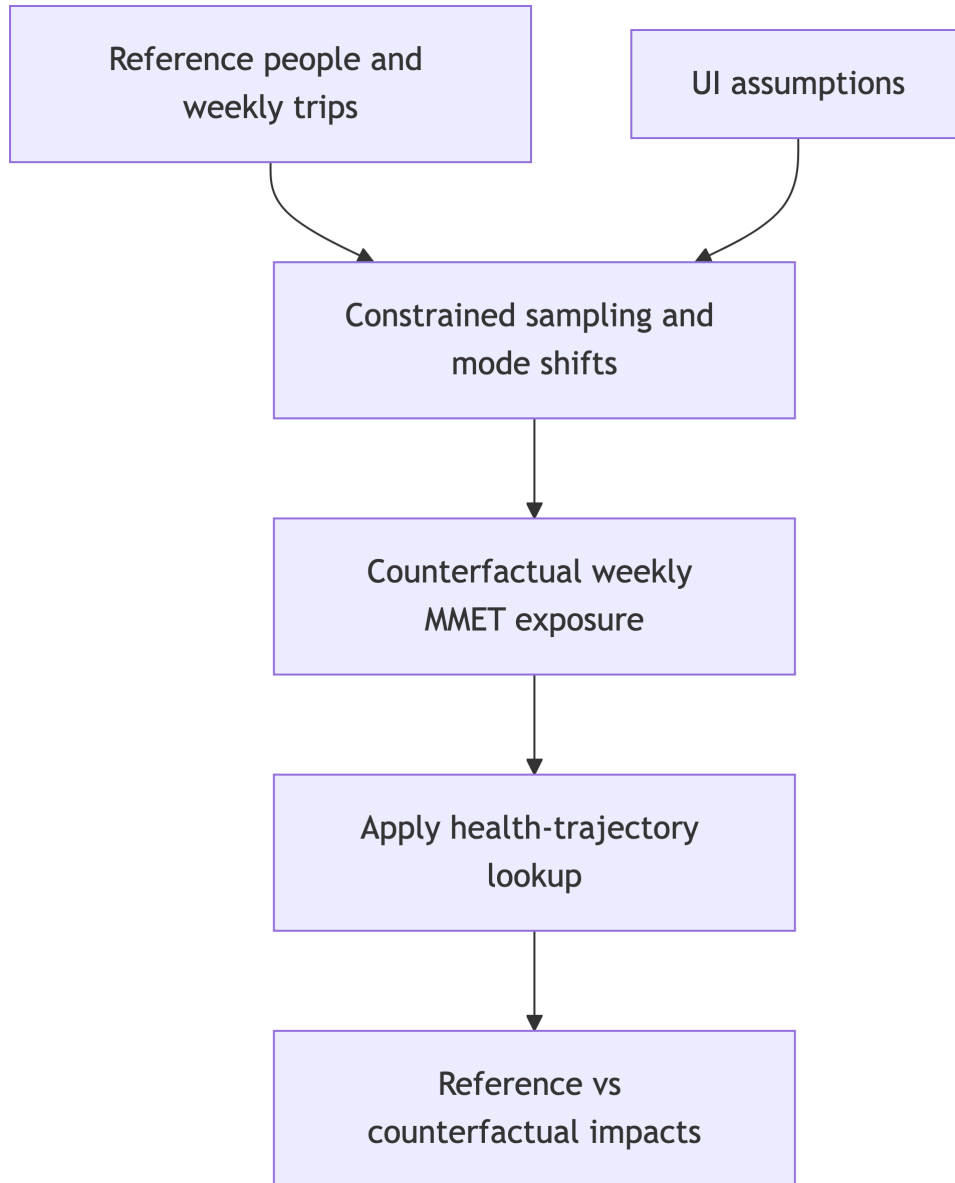
i Document status

This is a methods-facing description of the behavior implemented in MIAMA-HUB, not a final model specification. **Implemented** means that the behavior is present in the current code. **Approximate** means that it is implemented through probabilistic sampling and may not exactly reproduce a requested distribution. **Planned** identifies a UI option or methodological feature that is represented in the design but does not yet change modeled outcomes.

1 Purpose and scope

MIAMA estimates how a proposed intervention may change walking and cycling and how the resulting change in physical activity may alter population health over time. The current workflow is a comparative scenario model:

1. select a reference synthetic population and its observed weekly travel;
2. translate user-entered active-travel assumptions into a counterfactual copy of those people and trips;
3. convert changed walking and cycling time into marginal metabolic-equivalent exposure (MMETs);
4. apply the MMET change to health-model trajectories; and
5. report the difference between the reference (without-scheme) and counterfactual (with-scheme) scenarios.



The model is not a transport-demand model and does not create a new population. It reallocates active travel among the existing synthetic people and trips, with a limited mechanism for induced recreational trips. The validity of the results therefore depends on both the external assumptions supplied through the app and the representativeness of the donor people and trips available in the selected geography.

2 Reference synthetic population

2.1 Unit of representation

The synthetic population represents approximately 5% of the source population. Each person row is currently assigned a population factor of 20 when aggregate health results are presented. Individual

rows remain fixed between scenarios: investment in active travel changes behavior, not the number of residents.

The data contain two related levels:

- **Individual level:** one row per synthetic person, including geography, demographics, household characteristics, and weekly activity.
- **Trip level:** reported trips linked to people by `census_id`, including travel mode, purpose, distance, duration, and a trip weight.

Person and trip records are required for the most complete workflow. Trip rows contain person identifiers and selected attached attributes, but they do not replace the individual table: people without usable trip rows and person-level physical-activity measures still matter.

2.2 Key variables from active travel to physical activity

Domain	Principal fields	Role in the model
Identity and geography	<code>census_id</code> , <code>nts_id</code> , LAD and region identifiers	Select the appraisal population and link people, trips, and health trajectories.
Demographics	<code>age1year</code> , <code>female</code> , ethnicity, deprivation, limiting condition	Describe the reference population, constrain counterfactual sampling, and identify health-model strata.
Weekly activity	<code>walktime_wkhr</code> , <code>cycletime_wkhr</code> , <code>sport_wkhr</code>	Person-level weekly hours used to identify mode users and calculate MMET exposure.
Trip mode	<code>trip_mainmode</code> , walking and cycling component fields	Identify active trips and record mode switches.
Trip amount	raw and mode-specific distance and duration	Preserve plausible observed journeys and calculate added or removed active minutes.
Trip purpose	<code>trip_purpose</code> , derived <code>trip_utilitarian</code>	Distinguish travel needs from discretionary recreational travel.
Trip representation	<code>weight_tripXhh</code>	Weight reference trip summaries; treatment of weights in counterfactual row changes remains provisional.

Mode-user indicators are derived from positive weekly walking or cycling time, with trip evidence used where appropriate. Active-trip indicators use the mode-specific trip component fields and main-mode coding. HUB adds explicit `user_walk`, `user_bike`, `trip_activemode`, and `trip_utilitarian` fields to make these model currencies inspectable.

2.3 Weekly physical-activity exposure

The model uses the week as its base behavioral timeframe. Current conversion factors are configurable and equal to:

Activity	MMET per hour
Walking	2.5
Cycling	5.8
Other vigorous activity	7.0

For person-level changes, weekly hours are converted to MMET-week-hours and added to or subtracted from the person’s reference MMET value. For an independently specified trip target, active minutes are compared between reference and counterfactual trips, divided by 60, and multiplied by the mode-specific intensity. User-triggered trip changes are excluded from this second calculation because their exposure is already represented by the person-level activity change.

This separation prevents the same behavioral change from being counted once as weekly user activity and again as changed trip minutes.

3 From physical activity to health trajectories

MIAMA-HM provides reference person-cycle health outcomes and a piecewise lookup describing the change in each outcome per additional MMET. HUB does not rerun the underlying microsimulation. It applies the scenario logic to the processed health-model outputs.

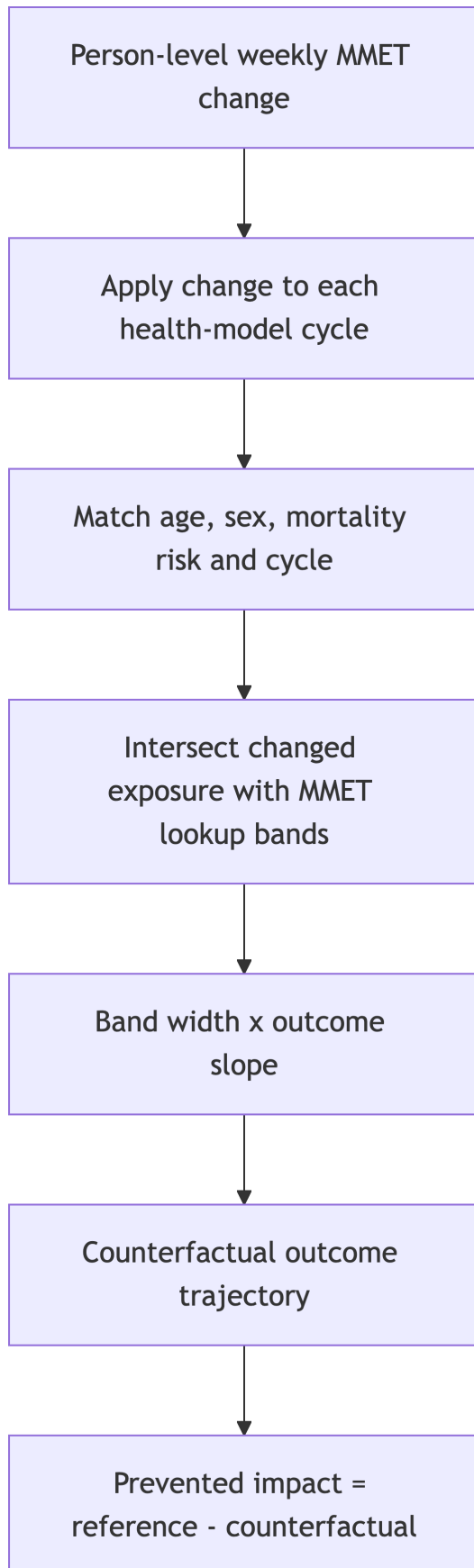
For each synthetic person:

$$\Delta MMET_i = MMET_{i,cf} - MMET_{i,ref}$$

Under the currently implemented **long-term** assumption, the same individual MMET change is applied in every modeled cycle:

$$MMET_{i,t,cf} = MMET_{i,t,ref} + \Delta MMET_i$$

The changed MMET interval is intersected with the applicable lookup bands for the person’s age, sex, mortality-risk decile, and model cycle. For each overlap, the model multiplies the interval width by the band-specific outcome slope and sums across bands. Counterfactual outcomes equal reference outcomes plus this technical delta.



The internal `d_*` fields use `counterfactual - reference`. Presentation fields reverse the sign so that a positive prevented value denotes a health gain. The configured assessment period is currently 40 years, excluding cycle 0 from impact totals. Population totals apply the synthetic-person factor described above.

The health outcome set includes mortality, life-year, and disease-incidence measures supplied by the death-share health-model artifacts. Effects attributed to walking or cycling are currently obtained after the combined health run by allocating impacts in proportion to each person’s signed mode-specific MMET contribution. This is an attribution of a combined nonlinear calculation, not an independent mode-specific rerun of the health model.

4 Ways to specify counterfactual active travel

4.1 Reference defaults

After geography and assessed modes are selected, HUB loads and filters the synthetic population and fills the UI profile’s reference `default_value` fields. It calculates reference values for all supported modes and common units up front; later UI changes can therefore switch displays without reloading the large data.

Reference values include user and trip counts, mode share, average distance and duration, and compact five-category distributions. Weekly values are canonical. Trip counts can be converted between day, week, and year; user-count conversion currently uses a factor of one because repeat-user behavior outside the observed week is not identified by the data.

4.2 Basic specification (Tab 2)

The basic interface offers alternative currencies for describing change:

Input route	Interpretation	Current modeling status
Number of mode users	Target number of weekly walking or cycling users	Implemented for walking and cycling.
Number of mode trips	Target weekly active-trip count, with UI timeframe and denominator conversion	Implemented for walking and cycling.
Distance or duration	Target aggregate or per-person active-travel amount	Planned: collected and summarized but not yet allocated to rows or health exposure.
Mode share	Target shares by travel mode	Planned: reference and UI values exist, but counterfactual shares do not yet drive trip sampling.

Only one basic input route is intended to be active at a time. User and trip targets use the same counterfactual machinery as the advanced interface; the advanced controls add information about who changes and which trips change.

4.3 Refined population specification (Tab 3)

For each assessed mode, Tab 3 can provide:

- a target number of affected mode users;
- a target mean age and proportion male;
- a target mean weekly PA exposure and proportion male within the PA display; and
- a UI option describing current versus new users.

Reference distributions are represented as a 5 x 2 table: five configured age or PA categories crossed with sex. The app displays the table as paired bars. Moving a mean or proportion slider does not transmit a new table to HUB. Instead, both UI and HUB reproduce it from the same reference bars and scalar counterfactual slider values.

For the numeric dimension, exponential tilting shifts the reference category percentages toward the requested mean while retaining the reference nonzero support. For the binary dimension, the requested proportion replaces the reference marginal. The two counterfactual marginals are combined using an outer product, which assumes independence between the numeric category and the binary grouping. If neither slider changes, the reference table is returned exactly.

Current configured categories are:

- age: 18–29, 30–39, 40–49, 50–59, and 60+;
- PA: sedentary, low, moderate, high, and very high, using provisional MMET cutoffs in configuration.

4.4 Refined trip specification (Tab 4)

For each assessed mode, Tab 4 can provide:

- a target number of weekly active trips;
- a target mean trip distance;
- a target proportion of utilitarian trips; and
- the percentage of new active-mode shifts expected to come from car.

Trip spread bars use five configured distance categories crossed with utilitarian/recreational purpose. The same slider redistribution method is used for display and for reconstructing internal counterfactual distance targets. The configured distance groups are 0–2, 2–5, 5–10, 10–30, and 30+ km.

5 Counterfactual sampling and redistribution

5.1 General design

Counterfactual data begin as a one-to-one copy of the filtered reference data. All changes are made only in that copy. The person count cannot increase; travel needs are preserved by switching trip mode rather than deleting utilitarian journeys. Sampling uses existing observed rows as donors so that joint characteristics not explicitly modeled remain plausible.

The random seed is fixed by the calling workflow, making a run reproducible for the same data and inputs. Sampling is without replacement when selecting people or existing trips. Induced trips and donor activity values may be sampled with replacement.

5.2 Changing the number of users

If the counterfactual target exceeds the reference user count, HUB samples existing non-users and marks them as **new_users**. If it is lower, HUB samples existing users and marks them as **ex_users**.

Candidate probabilities combine the available mode-specific Tab 3 constraints:

$$w_i = w_{sex,i} \times w_{age,i} \times w_{PA,i}.$$

The category percentages reconstructed from the sliders are used directly as relative sampling weights. They are not exact quotas and are not currently calibrated for the prevalence of each category in the candidate pool. If too few candidates have positive combined weight, all positive-weight candidates are selected first and the unavoidable remainder is sampled uniformly. The counterfactual report records this relaxation.

New users receive a weekly mode-activity value sampled with replacement from observed current users of that mode; the configured fallback is one hour per week if no donor value exists. Ex-users' mode activity is set to zero. This changes the person's MMET exposure.

When trips are available, new-user status also triggers mode shifts among that person's plausible non-active trips. The number is based on the configured weekly trips-per-user assumption; observed trip rates among current users are a fallback. For ex-users, their active trips are switched away from the active mode, currently to car.

Warning

The age and PA spread targets constrain **which candidates change status**. They do not impose exact distributions on all final counterfactual users. In addition, the PA constraint is based on candidate PA before newly sampled mode activity is added, so the final PA distribution of new users can differ from the displayed target.

5.3 Changing the number of trips

For an increase, 90% of the additional active trips are currently treated as mode shifts and 10% as induced recreational trips. These shares are configurable counterfactual assumptions.

Mode-shift candidates must be existing non-active utilitarian trips. Candidate distances are limited to 120% of the maximum observed reference active-mode distance to avoid implausible switches. Their sampling probabilities reflect the requested distance distribution or, when only a mean is available, a smooth preference for distances near that mean. The requested car-source share rebalances the car and non-car candidate pools while retaining the distance weights. The realized share can differ in finite or constrained samples and is reported.

Selected trips retain their observed raw distance and duration; the target active-mode fields are populated from those raw values. For induced travel, HUB copies observed active-trip donor rows, assigns new trip identifiers, marks them recreational, and assigns trip weight 1.

For a decrease, sampled active trips are switched to the configured alternative mode, currently car. Rows are not deleted because the underlying travel need is assumed to remain.

Warning

Although the Tab 4 utilitarian-proportion value is parsed and reported as a sampling constraint, it is not yet applied in `cf_trip_candidate_weights()`. At present, existing mode-shift candidates are all utilitarian and induced rows are all recreational, so the realized purpose split is mainly determined by the fixed 90/10 mechanism rather than the purpose slider.

5.4 Weights and represented totals

Reference trip summaries use `weight_tripXhh`. Counterfactual operations currently alter physical rows: shifted trips retain their existing weights and induced rows receive weight 1. Consequently, a requested numeric change and a weighted result are not guaranteed to coincide. This is acceptable for early mechanism testing but requires a settled weighting strategy before substantive appraisal use.

6 What each refinement currently controls

UI information	Stored/reconstructed quantity	Current effect on counterfactual data
Mode-specific user count	Weekly target users	Adds new-user or ex-user statuses among existing people.
Mean age + sex proportion	Five age categories and sex marginal	Weights candidate people changing status.
Mean PA + sex proportion	Five PA categories and sex marginal	Weights candidate people using their pre-change PA.

UI information	Stored/reconstructed quantity	Current effect on counterfactual data
Current/new-user percentage	Scalar profile value	Planned: parsed but not currently used to partition the target.
Mode-specific trip count	Weekly target trip rows	Shifts existing trips and adds induced recreational trips.
Mean trip distance	Five distance-category target or smooth mean preference	Weights plausible trip candidates.
Utilitarian proportion	Purpose marginal	Planned: parsed but not currently included in candidate weights.
Car diversion percentage	Expected car share among shifted trips	Rebalances car versus other source-mode candidate pools.
Basic distance/duration target	Aggregate exposure target	Planned: no allocation to people/trips.
Basic mode share target	Aggregate shares	Planned: no conversion to per-mode trip targets.

7 Validation, uncertainty, and remaining work

The current implementation makes assumptions visible in `counterfactual_report`, including requested and realized counts, active sampling constraints, fallback sampling, induced trips, diversion shares, and MMET components. This supports mechanical checking, but it is not a substitute for empirical validation.

7.1 Priority methodological TODOs

1. **Trip purpose targeting:** apply the Tab 4 utilitarian/recreational target in trip candidate selection and verify the realized split.
2. **Candidate-weight calibration:** decide whether category weights should be calibrated as target share divided by candidate prevalence, or replaced by exact/near-exact quota sampling. Test on large and sparse geographies.
3. **Final-distribution validation:** report requested versus realized age, sex, PA, distance, purpose, and source-mode distributions after all changes, not only the constraints supplied to the sampler.
4. **New versus current users:** implement the UI percentage explicitly and define whether spread controls describe new users, all affected users, or the complete counterfactual mode-user population.
5. **PA targeting semantics:** decide whether PA categories refer to pre-change activity, post-change total activity, or mode-specific added activity, then align sampling and UI wording.
6. **Trip weights:** define whether targets represent physical synthetic rows, weighted trips, or actual population trips; update shifts and induced trips accordingly.
7. **Basic distance/duration and mode share:** translate these inputs into operational user/trip targets or clearly mark them as non-modeling inputs.

8. **Short-term effects:** implement a time-limited intervention effect rather than applying the MMET change to every health cycle.
9. **Category and intensity review:** validate age, distance, PA cutoffs and MMET intensity factors against agreed evidence and document provenance.
10. **Mode coverage:** establish defensible e-bike and public-transport walking exposure rules; current person-level user changes support walking and cycling only.
11. **Uncertainty analysis:** quantify Monte Carlo variation from sampling and parameter uncertainty; a fixed seed currently provides reproducibility but not uncertainty intervals.
12. **External validation:** compare reference summaries, modeled behavioral changes, and health-response magnitudes with independent observed data and established appraisal tools.

7.2 Recommended checks for each appraisal

- Confirm that reference user/trip counts and spread plots are credible for the selected geography and are not using a sparse-data fallback.
- Compare requested and realized counterfactual changes in the report.
- Inspect the number of relaxed sampling selections and induced trip rows.
- Check the individual walking/cycling MMET components before reviewing health outcomes.
- Confirm sign conventions: internal delta is counterfactual minus reference; prevented outcomes are reference minus counterfactual.
- Treat very small samples and unsupported modes as development tests, not substantive estimates.

8 Implementation traceability

This description is based on the current implementation, principally:

- `R/reference_data_extract_reference_ui_values.R`
- `R/reference_data_spread_functions.R`
- `R/counterfactual_data_sampling_functions.R`
- `R/counterfactual_data_apply_ui_values.R`
- `R/counterfactual_data_apply_health_outcomes.R`
- `R/results_data_prepare.R`
- `R/config.R` and `R/constants.R`

The companion slide deck is `docs/methodology_slides.qmd`. The package README remains the developer-facing integration reference; where terminology differs, the implementation and the status statements in this methods document take precedence until the discrepancy is resolved.